COMPUTER FUNDAMENTALS

Unit-I: Introduction to Computers

Comprehensive Study Notes

Topic 1: What is a Computer?

1.1 Definition

A computer is an electronic device that processes data according to a set of instructions called a program. It accepts data as input, processes it, and produces output. It can perform complex calculations, store large amounts of data, and communicate with other computers globally.

1.2 Characteristics of Computers

Characteristic	Description
Speed	Computers can perform tasks in microseconds, nanoseconds, and even picoseconds. A modern computer can perform billions of operations per second.
Accuracy	Computers are highly accurate. Errors occur only due to human mistakes or faulty data.
Diligence	Computers can work for hours without getting tired or losing concentration.
Versatility	Computers can perform different types of tasks simultaneously.
Storage	Computers can store large amounts of data permanently.
Automation	Computers can work automatically once instructions are given.
No Feelings	Computers do not have emotions like humans.
Reliability	Computers are highly reliable and consistent in their performance.
Communication	Computers can connect and communicate with other computers globally.

1.3 Capabilities of Computers

- Can perform complex calculations rapidly
- Can process large volumes of data
- Can store and retrieve information
- Can automate repetitive tasks
- Can connect and communicate globally
- Can display graphics and multimedia
- Can assist in decision making
- Can perform multiple tasks simultaneously

1.4 Limitations of Computers

- Cannot think or reason independently
- Cannot make decisions on their own
- Cannot understand emotions or feelings
- Cannot correct mistakes on their own
- Dependent on human input and instructions
- Cannot learn from experience
- Cannot handle unexpected situations
- Cannot replace human creativity

Topic 2: Evolution of Computers

2.1 Historical Timeline

Period	Device	Significance
3000 BC	Abacus	First calculating device
1642	Pascaline	Mechanical calculator by Blaise Pascal
1671	Leibniz Calculator	Stepped reckoner
1822	Difference Engine	Charles Babbage - Father of Computers
1834	Analytical Engine	First general-purpose computer concept
1937	Mark I	First electro-mechanical computer
1945	ENIAC	First electronic computer

Key Contributions

Contributor	Contribution
Charles Babbage	Known as Father of Computers; designed Difference Engine (1822) and Analytical Engine (1834)
Ada Lovelace	First programmer; wrote algorithms for Analytical Engine
Alan Turing	Developed the concept of the Universal Machine (Turing Machine)
John von Neumann	Proposed the stored-program concept (Von Neumann Architecture)

Topic 3: Generations of Computers

The term "generation" in computer technology refers to a change in the underlying technology used to build a computer. There are five generations:

3.1 Generations Summary

Generation	Period	Technology	Examples
First	1940-1956	Vacuum Tubes	ENIAC, EDVAC, UNIVAC
Second	1956-1963	Transistors	IBM 7090, IBM 7094, CDC 1604
Third	1964-1971	Integrated Circuits (ICs)	IBM 360 series, PDP-8, PDP-11
Fourth	1971-1980	Microprocessors (VLSI)	IBM PC, Apple II, Macintosh
Fifth	1980-Present	AI, Parallel Processing (ULSI)	Modern PCs, laptops, smartphones

3.2 First Generation (1940-1956)

Feature	Description
Technology	Vacuum tubes
Characteristics	Very large in size, consumed high power, generated heat, expensive, unreliable
Memory	Magnetic drums
Input/Output	Punch cards and paper tape
Languages	Machine language

Examples	ENIAC, EDVAC, UNIVAC

3.3 Second Generation (1956-1963)

Feature	Description
Technology	Transistors
Characteristics	Smaller size, less power consumption, more reliable, faster
Memory	Magnetic core
Input/Output	Magnetic tape and disk
Languages	Assembly language
Examples	IBM 7090, IBM 7094, CDC 1604

3.4 Third Generation (1964-1971)

Feature	Description
Technology	Integrated Circuits (ICs)
Characteristics	Even smaller, more powerful, low cost, multitasking
Memory	Semiconductor memory
Input/Output	Keyboard and monitor
Languages	High-level languages (COBOL, FORTRAN, BASIC)
Examples	IBM 360 series, PDP-8, PDP-11

3.5 Fourth Generation (1971-1980)

Feature	Description

Technology	Microprocessors (VLSI - Very Large-Scale Integration)
Characteristics	Personal computers, GUI (Graphical User Interface), networking
Memory	RAM and ROM
Input/Output	Keyboard, mouse, monitor, printer
Languages	High-level languages, C, C++, Pascal
Examples	IBM PC, Apple II, Macintosh

3.6 Fifth Generation (1980-Present)

Feature	Description
Technology	AI (Artificial Intelligence), Parallel Processing, ULSI (Ultra Large-Scale Integration)
Characteristics	High-speed, AI applications, robotics, quantum computing
Memory	Cache memory, virtual memory
Input/Output	Voice recognition, touch screen
Languages	Prolog, LISP, advanced programming languages
Examples	Modern PCs, laptops, smartphones, supercomputers

Topic 4: Types of Computers

4.1 Classification by Data Processing Type

Type	Description	Examples
Analog Computers	Process continuous data (physical quantities like temperature, pressure, voltage)	Thermometer, Speedometer, Analog clock
Digital Computers	Process data in binary form (0 and 1). Most common type	PCs, Laptops, Smartphones
Hybrid Computers	Combination of analog and digital computers. Used in specialized applications	Medical equipment, Industrial control

4.2 Classification by Size and Power

Type	Size	Power	Usage
Supercomputers	Largest	Very High	Weather, defence, research
Mainframe Computers	Large	High	Banking, large organizations
Minicomputers	Medium	Moderate	Universities, small companies
Microcomputers	Small	Low-Moderate	Personal use, offices

Supercomputers

The most powerful and expensive computers.

- Used for complex calculations, weather forecasting, nuclear research, space exploration
- Speed measured in FLOPS (Floating-point Operations Per Second)

- Examples: Cray, IBM Blue Gene, Fugaku

Mainframe Computers

Large, powerful systems used by large organizations.

- Used for bulk data processing
- Can support many users simultaneously (hundreds to thousands)
- Used in banking, insurance, government, airlines
- Examples: IBM zSeries, Unisys

Minicomputers

Medium-sized computers used by small organizations for specific purposes.

- Applications: Inventory management, accounting, research
- Examples: DEC PDP-11, IBM AS/400

Microcomputers

Also known as Personal Computers (PCs).

- Used for word processing, spreadsheets, internet, gaming
- Types: Desktop PCs, Laptops, Tablets, Smartphones
- Examples: Dell, HP, Apple MacBook, Samsung Galaxy

4.3 Classification by Purpose

Type	Description	Examples
General Purpose Computers	Used for various tasks	PCs, Laptops
Special Purpose Computers	Used for specific tasks	Embedded systems, ATMs, calculators

Topic 5: Block Diagram of Computer

The block diagram represents the organization of a computer system. It consists of four main units:

```
            +-------------------+
            |   INPUT UNIT      |
            +---------+---------+
                      |
                      v
      +---------------+---------------+
      |   CENTRAL PROCESSING UNIT   |
      |  (CPU)                        |
      |  +--------+  +-----------+    |
      |  |  ALU   |  |     CU    |    |
      |  +--------+  +-----------+    |
      |  +----------------------+     |
      |  |      REGISTERS       |     |
      |  | (AC, PC, IR, MAR)    |     |
      |  +----------------------+     |
      +---------------+---------------+
                      |
      +---------------+---------------+
      |       MEMORY UNIT           |
      |  (RAM, ROM, Cache)          |
      +---------------+---------------+
                      |
                      v
            +-------------------+
            |   OUTPUT UNIT     |
            +-------------------+
                      |
                      v
            +-------------------+
            |  SECONDARY        |
            |  STORAGE          |
            +-------------------+
```

5.1 Input Unit

Allows users to enter data and instructions.

Functions:

- Accepts data from the outside world
- Converts data into a form understandable by the computer
- Supplies converted data to the computer system

Examples:

Keyboard, Mouse, Scanner, Microphone, Joystick, Trackball

5.2 Central Processing Unit (CPU)

The brain or heart of the computer. It processes all data and instructions.

a. Arithmetic Logic Unit (ALU)

- Performs arithmetic operations: Addition, Subtraction, Multiplication, Division
- Performs logical operations: AND, OR, NOT, comparisons
- All operations are performed on binary data

b. Control Unit (CU)

- Controls and coordinates all operations
- Fetches instructions from memory
- Decodes instructions
- Executes instructions
- Controls the flow of data
- Generates control signals

c. Registers

Small, high-speed storage locations within the CPU.

Register	Full Form	Function
AC	Accumulator	Stores results of operations
PC	Program Counter	Holds address of next instruction

IR	Instruction Register	Holds current instruction
MAR	Memory Address Register	Holds address of memory location
MDR	Memory Data Register	Holds data to be stored or retrieved

d. Cache Memory

- Fast memory between CPU and RAM
- Stores frequently accessed data
- Improves processing speed
- Uses SRAM technology

5.3 Memory Unit

Stores data, instructions, and results.

Primary Memory:

- RAM (Random Access Memory) - Volatile memory
- ROM (Read Only Memory) - Non-volatile memory

Secondary Memory:

- Hard Disk Drive (HDD)
- CD, DVD
- USB Drives
- Solid State Drives (SSD)

5.4 Output Unit

Displays or prints results from the computer.

Functions:

- Accepts results from the computer
- Converts results into human-readable form
- Presents results to the user

Examples:

Monitor, Printer, Speaker, Projector

Topic 6: Basic Components of Computer System

6.1 Input Devices

Device	Description
Keyboard	Most common input device
Mouse	Pointing device
Scanner	Converts paper documents to digital
Microphone	Audio input
Joystick	Gaming input
Trackball	Alternative pointing device
Touch Screen	Direct touch input
Digital Camera	Image input

6.2 Output Devices

Device	Description
Monitor (VDU)	Visual display
Printer	Hard copy output
Speaker	Audio output
Projector	Large display
Plotter	Large format printing

6.3 Types of Memory

Category	Type	Volatile	Examples
Primary Memory	RAM	Yes (Volatile)	SRAM, DRAM
	ROM	No (Non-Volatile)	MROM, PROM, EPROM, EEPROM
Secondary Memory		No (Non-Volatile)	HDD, SSD, CD, DVD, USB

6.4 Processor Speed

Unit	Full Form	Description
Hz/MHz/GHz	Hertz/Megahertz/Gigahertz	Clock speed - number of cycles per second
MIPS	Millions of Instructions Per Second	Number of instructions executed per second
FLOPS	Floating-point Operations Per Second	Used for supercomputers

6.5 Types of Processors

Manufacturer	Examples
Intel	Pentium, Core i3, Core i5, Core i7, Core i9
AMD	Athlon, Ryzen
ARM	Used in mobile devices

6.6 Instruction Set Architecture

Architecture	Full Form	Description

RISC	Reduced Instruction Set Computer	Simple, limited instructions
CISC	Complex Instruction Set Computer	Complex, multiple instructions

Topic 7: Quick Revision

7.1 Characteristics of Computers

- **Speed** - Billion operations per second
- **Accuracy** - 100% accurate
- **Diligence** - Never gets tired
- **Versatility** - Different tasks
- **Storage** - Large data storage
- **Automation** - Works automatically
- **No Feelings** - No emotions
- **Reliability** - Consistent performance
- **Communication** - Global connectivity

7.2 Generations Summary

Generation	Technology	Period
1st	Vacuum Tubes	1940-1956
2nd	Transistors	1956-1963
3rd	Integrated Circuits (ICs)	1964-1971
4th	Microprocessors (VLSI)	1971-1980
5th	AI, Parallel Processing (ULSI)	1980-Present

7.3 Block Diagram Flow

```
Input Unit  -->  CPU (ALU + CU + Registers)  -->  Output Unit
                              |
                         Memory Unit
```

```
                |
      Secondary Storage
```

7.4 Important Full Forms

Abbreviation	Full Form
ENIAC	Electronic Numerical Integrator and Computer
UNIVAC	Universal Automatic Computer
ALU	Arithmetic Logic Unit
CU	Control Unit
CPU	Central Processing Unit
RAM	Random Access Memory
ROM	Read Only Memory
PROM	Programmable ROM
EPROM	Erasable PROM
EEPROM	Electrically Erasable PROM
VLSI	Very Large Scale Integration
ULSI	Ultra Large Scale Integration
MIPS	Millions of Instructions Per Second
FLOPS	Floating-point Operations Per Second

Questions

Short Answer Questions (2-5 Marks)

1. What is a computer? Write its characteristics.
2. What are the limitations of computers?
3. Define "Generation of Computers".
4. What is the difference between analog, digital, and hybrid computers?
5. Explain the block diagram of a computer.
6. What is the difference between ALU and CU?
7. Define registers. What is their purpose?
8. What is processor speed? How is it measured?
9. What are the types of computers based on size?
10. What is an instruction set?
11. What is the difference between microcomputers and minicomputers?
12. What are supercomputers? Give examples.
13. What is the full form of ENIAC?
14. What are the advantages of transistors over vacuum tubes?
15. What is the role of memory unit?
16. What is the difference between RAM and ROM?
17. What is VLSI technology?
18. Explain the Von Neumann architecture.
19. What is cache memory? Why is it used?
20. What is the difference between primary and secondary memory?

Long Answer Questions (10 Marks)

1. Explain the characteristics and limitations of computers in detail.
2. Describe the evolution of computers from Abacus to modern computers.
3. Explain the generations of computers with examples and characteristics.
4. Draw and explain the block diagram of a computer system in detail.
5. Classify computers by size and power. Explain each type in detail.
6. Explain CPU components in detail with a neat diagram.
7. What are the different types of processors? Explain with examples.

8. What is the difference between RISC and CISC architecture?
9. Explain the memory hierarchy in a computer system.
10. Write short notes on: (a) Cache Memory (b) Registers (c) ALU (d) Control Unit

Objective Questions (1 Mark)

1. Father of Computers: (a) Pascal (b) Babbage (c) Turing (d) von Neumann
   Answer: (b)
2. ENIAC developed in: (a) 1945 (b) 1950 (c) 1960 (d) 1970
   Answer: (a)
3. 1st Generation used: (a) Transistors (b) Vacuum Tubes (c) ICs (d) Microprocessors
   Answer: (b)
4. Brain of computer: (a) Hard Disk (b) Memory (c) CPU (d) Input Unit
   Answer: (c)
5. Full form of ALU: (a) Arithmetic Logic Unit (b) Arithmetic Language Unit (c) Both
   Answer: (a)
6. Used for weather forecasting: (a) Micro (b) Mainframe (c) Supercomputer (d) Mini
   Answer: (c)
7. Microprocessors introduced in: (a) 2nd (b) 3rd (c) 4th (d) 5th Generation
   Answer: (c)
8. Full form of VLSI: (a) Very Large Scale Integration (b) Very Low Scale (c) Virtual Large
   Answer: (a)
9. Which is volatile memory? (a) ROM (b) Hard Disk (c) RAM (d) CD
   Answer: (c)
10. Which generation introduced ICs? (a) 1st (b) 2nd (c) 3rd (d) 4th
    Answer: (c)
11. Hybrid computers are combination of: (a) Digital + Analog (b) Digital + Super (c) Analog + Mainframe (d) None
    Answer: (a)
12. Which memory is used as cache? (a) SRAM (b) DRAM (c) ROM (d) Hard Disk
    Answer: (a)
13. Speed of processor is measured in: (a) MHz/GHz (b) MBPS (c) GBPS (d) None
    Answer: (a)
14. Which is not a type of computer by size? (a) Micro (b) Mini (c) Mainframe (d) Medium
    Answer: (d)

End of Unit-I Notes